

SOLVED PROBLEM 2. The resistance of decinormal solution of a salt occupying a volume between two platinum electrodes 1.80 cm apart and 5.4 cm² in area was found to be 32 ohms. Calculate the equivalent conductance of the solution.

SOLUTION

Here

$$l = 1.80 \text{ cm and } A = 5.4$$

$$\therefore \text{ cell constant } x = \frac{l}{A} = \frac{1.80}{5.4} = \frac{1}{3}$$

$$\text{observed conductance} = \frac{1}{32} \text{ mhos}$$

Since the solution is *N*/10, $V = 10,000 \text{ ml}$

Now, specific conductance = $x \times \text{obs. conductance}$

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or

$$\kappa = \frac{1}{3} \times \frac{1}{32} = \frac{1}{96} \text{ mhos}$$

$$\Lambda = \kappa \times V$$

$$= \frac{1}{96} \times 10,000$$

$$= 104.1 \text{ mhos cm}^2 \text{ equiv}^{-1}$$

SOLVED PROBLEM 1. A solution of silver nitrate containing 12.14 g of silver in 50 ml of solution was electrolysed between platinum electrodes. After electrolysis, 50 ml of the anode solution was found to contain 11.55 g of silver, while 1.25 g of metallic silver was deposited on the cathode. Calculate the transport number of Ag^+ and NO_3^- ions.

SOLUTION

Weight of Ag in 50 ml of the solution before electrolysis = 12.14 g

Weight of Ag in 50 ml of the solution after electrolysis = 11.55 g

$\therefore$ Fall in concentration of Ag = 12.14 – 11.55

$$= 0.59 \text{ g}$$

$$= 0.0055 \text{ g eq}$$

Weight of Ag deposited in silver coulometer = 1.25 g

$$= \frac{1.25}{108} \text{ g eq}$$

$$= 0.0116$$

$$\begin{aligned} \text{Hence, transport number of } \text{Ag}^+ (t_{\text{Ag}^+}) &= \frac{\text{Fall in conc. around anode}}{\text{No. of g eqvt deposited in silver coulometer}} \\ &= \frac{0.0055}{0.0116} \\ &= 0.474 \end{aligned}$$

$$\begin{aligned} \text{and Transport number of } \text{NO}_3^- \text{ ion } (t_{\text{NO}_3^-}) &= 1 - 0.474 \\ &= 0.526 \end{aligned}$$

SOLVED PROBLEM 2. In an electrolysis of copper sulphate between copper electrodes the total mass of copper deposited at the cathode was 0.153 g and the masses of copper per unit volume of the anode liquid before and after electrolysis were 0.79 and 0.91 g respectively. Calculate the transport number of the Cu^{2+} and SO_4^{2-} ions.

SOLUTION

Wt. of copper in the anode liquid before electrolysis = 0.79 g

Wt. of copper in the anode liquid after electrolysis = 0.91 g

Increase in weight = 0.91 – 0.79

$$= 0.12 \text{ g}$$

Increase in weight of copper cathode in the coulometer = 0.153 g

This means that if no copper had migrated from the anode, increase in weight would have been 0.153 g.

But actual increase = 0.12

Fall in concentration due to migration of Cu^{2+} ions = 0.153 – 0.12

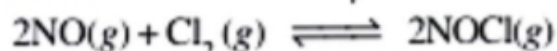
$$= 0.033$$

$$\begin{aligned} \therefore \text{Transport number of } \text{Cu}^{2+} \text{ ion} &= \frac{0.033}{0.153} \\ &= 0.215 \end{aligned}$$

$$\begin{aligned} \text{and Transport number of } \text{SO}_4^{2-} \text{ ion} &= (1 - 0.215) \\ &= 0.785 \end{aligned}$$

$$= 1.9 \times 10^3$$

SOLVED PROBLEM 2. The value of K_p at 25°C for the reaction



is $1.9 \times 10^3 \text{ atm}^{-1}$. Calculate the value of K_c at the same temperature.

SOLUTION

We can write the general expression as

$$K_p = K_c (RT)^{\Delta n} \text{ or } K_c = \frac{K_p}{(RT)^{\Delta n}}$$

Here,

$$T = 25 + 273 = 298 \text{ K}$$

$$R = 0.0821$$

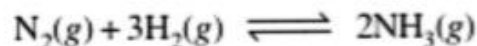
$$\Delta n = 2 - (2 + 1) = -1$$

$$K_p = 1.9 \times 10^3$$

Substituting these values in the general expression

$$\begin{aligned} K_c &= \frac{1.9 \times 10^3}{(0.0821 \times 298)^{-1}} \\ &= 4.6 \times 10^4 \end{aligned}$$

SOLVED PROBLEM 2. Some nitrogen and hydrogen gases are pumped into an empty five-litre glass bulb at 500°C. When equilibrium is established, 3.00 moles of N_2 , 2.10 moles of H_2 and 0.298 mole of NH_3 are found to be present. Find the value of K_c for the reaction



at 500°C.

SOLUTION

The equilibrium concentrations are obtained by dividing the number of moles of each reactant and product by the volume, 5.00 litres. Thus,

$$[N_2] = 3.00 \text{ mole}/5.00 \text{ L} = 0.600 \text{ M}$$

$$[H_2] = 2.10 \text{ mole}/5.00 \text{ L} = 0.420 \text{ M}$$

$$[NH_3] = 0.298 \text{ mole}/5.00 \text{ L} = 0.0596 \text{ M}$$

Substituting these concentrations (not number of moles) in the equilibrium constant expression, we get the value of K_c .

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{(0.0596)^2}{(0.600)(0.420)^3} = 0.080$$

Thus, for the reaction of H_2 and N_2 to form NH_3 at 500°C, we can write

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = \mathbf{0.080}$$